

Predictions



- 1 a Interpolation b Extrapolation c Interpolation d Extrapolation

- 2 a $y = -23.6079$ b $y = 67.8912$ c $y = 36.056$ d $y = 8.7592$

- 3 50.536 cm

- 4 a $y = 0.5x + 5$ b 8.45

- 5 a $y = -1.2x + 16$ b $y = 8$

- 6 a $y = 13.44$
b The data is strongly correlated and the prediction is an interpolation, so the prediction is reliable.

- 7 a $y = 32.16$
b The data is weakly correlated and the prediction is an extrapolation, so the prediction is unreliable.

- 8 a $y = 38.53$
b The data is moderately correlated and the prediction is an extrapolation far from the data range. There's no telling that the linear trend continues so the prediction is unreliable.

- 9 a $y = 94.04$
b The data is moderately correlated and the prediction is an interpolation, so the prediction is moderately reliable.

- 10 a $y = -45.69$
b The data is strongly correlated and the prediction is an extrapolation close to the data range, so the prediction is reliable.



Applications

11

- a Yes, because it is interpolation. b 2448 Pa
c
i Yes ii No iii No iv No

12

- a 90 b 22°C
c
i No ii Yes iii No iv No

13

- a 3 seconds b 2.5 hours

14

- a 20 years b 1300 fish c 11 years

15

- a 8500 m b 6500 m c 300 seconds d 1800 seconds

16

- a 20°C b 210 chirps per minute
c 130 chirps per minute

17

- a 48 m b $y = 3x + 30$
c 52.5 m d Yes, because it is using interpolation.

18

- a $y = -0.005x + 49$ b $y = 29^{\circ}\text{C}$
c Yes, because it is using interpolation.

19

- a Negative gradient b -2.5
c $y = -2.5x + 105$ d $y = 95$

20

- a $\frac{1}{16}$ b $c = 100$ c \$400 000

21

a $A = -25T + 750$

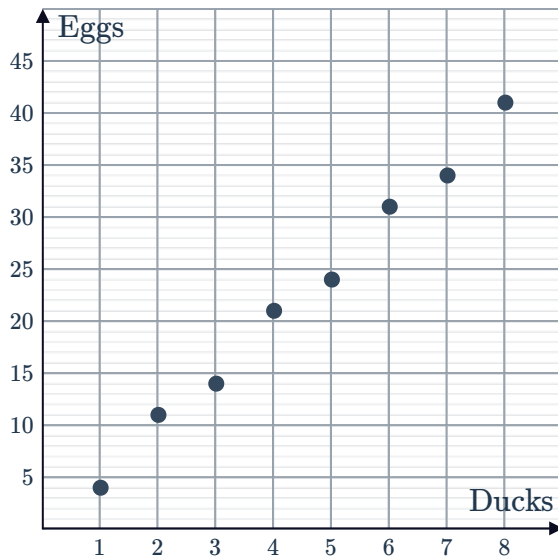
b $A = 100$ hectares

c No, because it is using extrapolation.

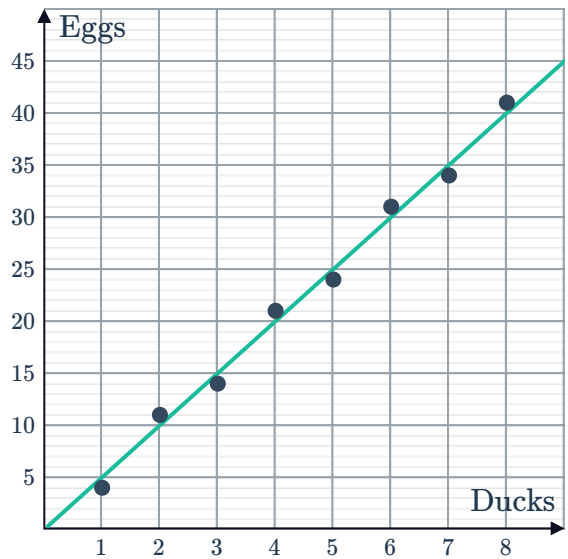


22

a



b



c 5

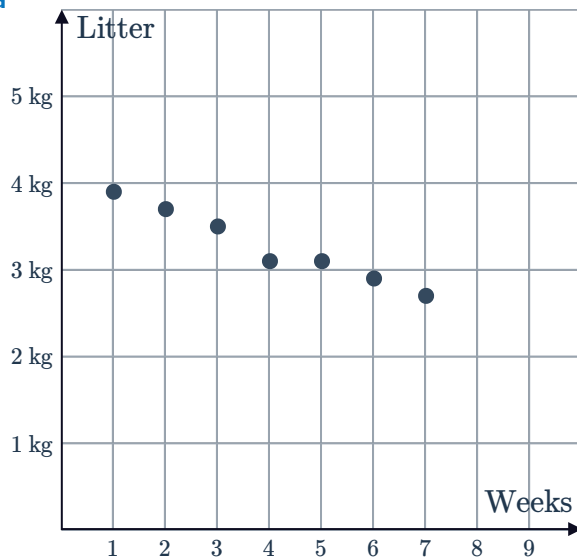
d 0

e $y = 5x$

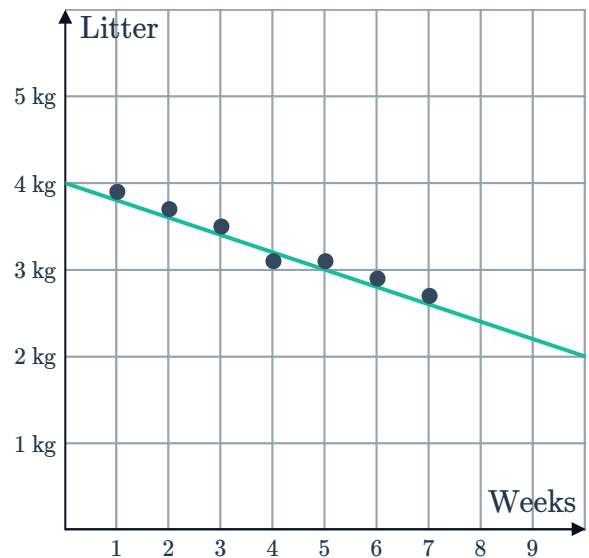
f 150 eggs

23

a



b Example answer:



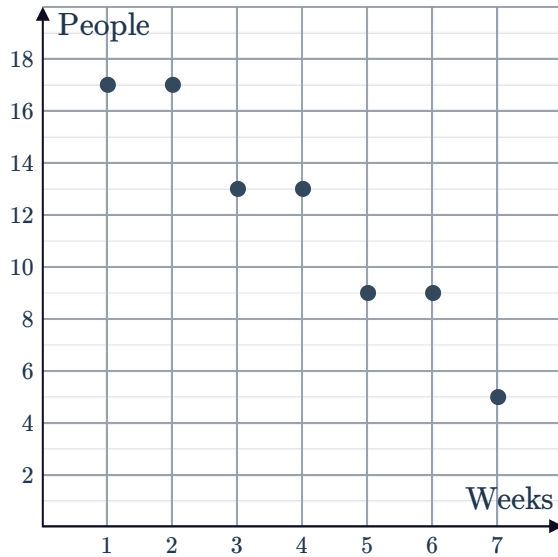
c Example answer: $y = -0.2x + 4$

d Approximately 2.4 kg

e No, because it is using extrapolation.

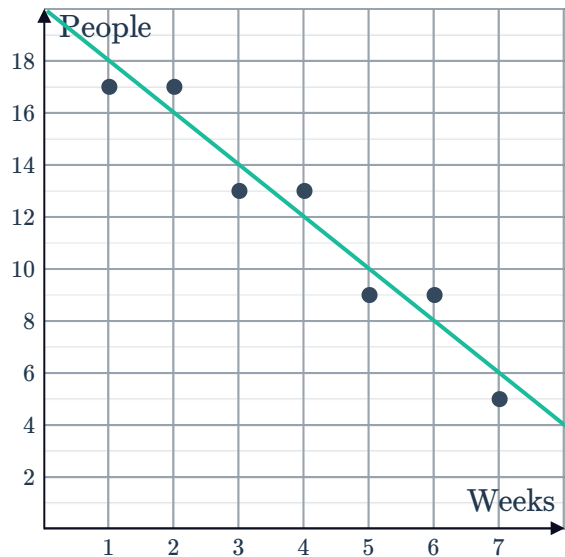
24

a



b

Example answer:



c Example answer: $y = -2x + 20$

d Approximately 0

25

- a 14 811
- b The y -intercept indicates that when a house is brand new, its value is, on average, \$14 811.
- c No. When a house is zero years old its age is not within the data range.
- d Approximately 4.9 years.
- e \$1 200 000

26

- a The y -intercept indicates that when a student watched no TV, their average mark in the exam, was 90.
- b Yes. When a student watches 0 hours, their time is within the data range, and the mark makes sense.
- c No

27

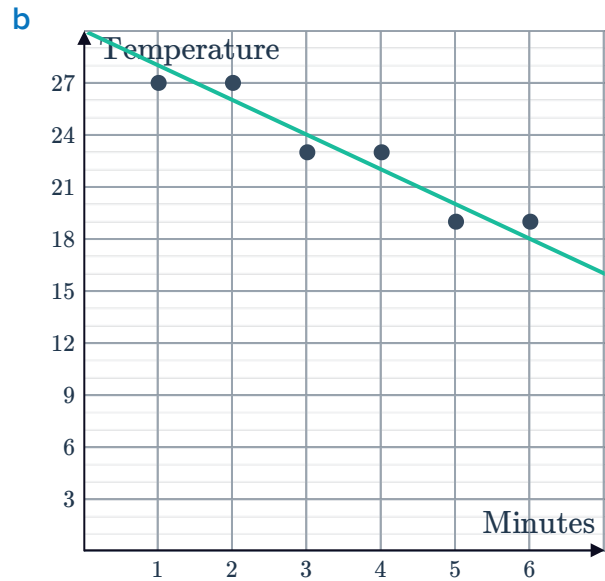
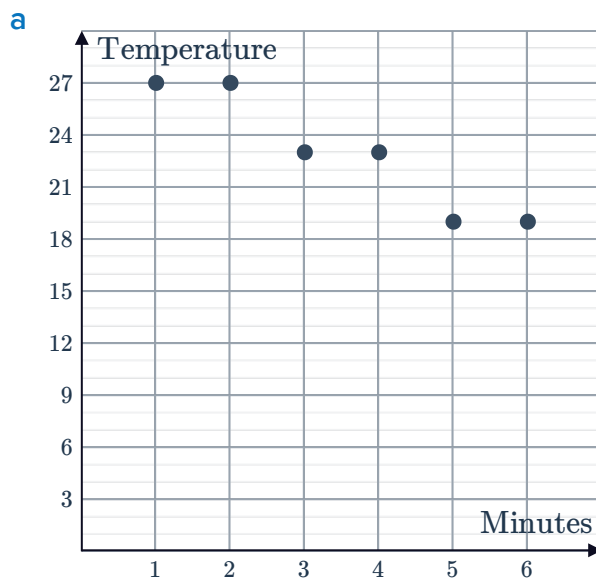
- a 18 000
- b The y -intercept indicates that when a Mitsubishi Lancer is brand new, its value is, on average, \$18 000 .
- c Yes. When a car is zero years old its age is within the data range.
- d 11 years
- e No. When a car is older than 11 years, the line goes below the x -axis, which would give a negative value, which is not possible. We cannot use the line to find the value since this age is not within the data range.

a $y = -3x + 60$



b 24

29



c -2

e $y = -2x + 30$

d 30

f $x = \frac{15}{2}$

30

- a For each 1000 km increase in distance driven, the least-square line predicts that the maintenance costs increase by \$990.
- b A car that is not driven at all is predicted to cost \$14620 to maintain.
- c No, because for a car that hasn't been driven, it should have little to no cost to maintain.
- d \$50 260
- e Yes, because it is an extrapolation close to the data range where the cost is likely to increase along with distance driven, and the correlation coefficient is high.
- f Yes, a car that is driven more will have more wear and tear and would be likely to have greater maintenance costs.